Intelligent Tutoring System

MorseMate: Morse Code Tutor

We developed a simple tutor that helps learners practice eight basic Morse-code letters. We use previous student errors to select a teaching strategy and give personalized feedback.

3. Define the case base

We stored five simulated past cases as JSON objects in **case_base.json**. Each case has a **problem** containing the letter, student answer, error type and confidence, and a **solution** containing the teaching strategy, feedback and practice task.

Case	Letter / answer	Error / result	Confidence	Stored strategy
C001	A / -.	Reversed order	Low	Read left to right
C002	S / ..	Missing symbol	Low	Count symbols
C003	E / -	Wrong symbols	Medium	Compare symbols
C004	N / -..	Extra symbol	Medium	Count symbols
C005	O / ---	Correct answer	High	Reinforce recall

4. Implement similarity assessment

We compare the new problem with every stored case using weighted attribute matching. We give **60 points** for the same error type, **25 points** for the same letter and **15 points** for the same confidence level. Each match is 1 if true and 0 otherwise.

Similarity = 60(same error) + 25(same letter) + 15(same confidence)

For our new case, a student enters **'.-'** for the letter **N** with **medium confidence**. The correct code is **'-.'**, so we identify the error as reversed order.

Rank	Case	Matching attributes	Similarity
1	C001	Error type	60%
2	C004	Letter and confidence	40%
3	C003	Confidence	15%
4	C002	None	0%
5	C005	None	0%

We retrieve **C001** because it has the highest similarity score. Although it concerns the letter A, both students reversed the symbols. If scores are equal, we use the case stored earlier.

5. Adapt the solution

We reuse C001's reading-order strategy. We change its example from A, **'.-'** or **dot then dash**, to N, **'-.'** or **dash then dot**. We also adjust the guidance to the student's medium confidence.

Our adapted feedback:

> Your answer: '.-'. Correct Morse code for N: '-.'. You reversed the order. Read left to right: dash then dot. Say the pattern aloud, then try once without looking.

Our practice task: Write N (-.) three times, keeping the same order.

6. Demonstrate learning: retain stage

We save the new problem and its adapted solution to **case_base.json** as **C006**. Our program confirms the save and displays the updated case list:

New case saved: C006
Updated case list (6 cases):

Case	Letter	Student answer	Error / result	Confidence
C001	A	-.	Reversed order	Low
C002	S	..	Missing symbol	Low
C003	E	-	Wrong symbols	Medium
C004	N	-..	Extra symbol	Medium
C005	O	---	Correct answer	High
C006	N	.-	Reversed order	Medium

We retain the letter **N**, answer **'.-'**, error type **reversed** and confidence **medium**, together with the **read_order** strategy, personalized feedback and practice task.

When we repeat the same student attempt, C006 receives **100% similarity** because all three compared attributes match. We can now retrieve this saved case for future feedback.